

g0-sense — Design Document

ai-ee v1 pipeline

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1 Board and Run Metadata

Field	Value
board	g0-sense
workspace	boards/g0-sense
phase	P2
gate status	no gates recorded yet
state created	2026-08-27T06:35:38
state updated	2026-08-27T08:42:51

2 Overview

Brief: g0-sense (unattended container run, 2026-08-27)

A small 2-layer USB-C powered temperature/humidity sensor node - a product-scope board (no mode token: design normally), meant to be sent to JLCPCB as-is.

- Power: USB-C receptacle used for POWER ONLY (5 V VBUS; 5.1 k CC pull-downs so any USB-C source or a legacy A-to-C cable enables VBUS; D+/D- unconnected). Sensible input protection for a USB-powered device (VBUS TVS, resettable fuse or equivalent, reverse/inrush as the architect sees fit).
- 3.3 V rail from a fixed LDO, ≥ 300 mA, JLC Basic/Preferred part if one fits (AMS1117-3.3 class is fine), with proper input/output capacitors.
- MCU: STM32G030F6P6 (TSSOP-20) on the internal oscillator - no crystal. Decoupling per the datasheet, NRST with a push button, BOOT0 handled.

- Sensor: Sensirion SHT40 (or the SHT4x family member in JLC stock) on I2C with pull-ups; give it a thermal isolation slot/cutout or at least keep it away from the LDO and MCU heat.
- Interfaces: 4-pin SWD header (0.1 in), 4-pin UART header (0.1 in: GND, 3V3, TX, RX) for readout/logging, one Qwiic/STEMMA-QT connector (JST SH 1.0 mm 4-pin, GND/3V3/SDA/SCL) sharing the sensor I2C bus.
- One user LED (GPIO, series resistor) and one power LED.
- 2 layers, 1.6 mm FR-4, HASL, green; JLC economy PCBA for the SMD parts, prototype qty 5. Prefer JLC Basic parts; Extended parts only where the brief names the part (MCU, sensor, connectors).
- Size: whatever an honest layout wants, roughly 35 x 25 mm or smaller; four M2 mounting holes if they fit without hurting the layout.
- No battery, no mains, no USB data, no RF.

Stackup

Chosen stackup: JLC2313_1.6 (reference/stackups.yaml, available: true)

3 Requirements

requirements - g0-sense

P0 requirements distilled from `brief/brief.md` (2026-08-27, unattended container run).

1. Function

Mode declaration: the brief carries NO mode token, so mode = none - design normally. Scope = product-scope per the brief's own words ("a product-scope board ... meant to be sent to JLCPCB as-is"): protection, filtering, connectors, thermal and enclosure-fit are all in scope, and their absence is a reviewer finding. Binding: the stated ~35 x 25 mm size is a SOFT preference, not a hard cap (see section 5). Stage under study: none.

The board is a small 2-layer USB-C powered temperature/humidity sensor node. A USB-C receptacle supplies 5 V (power only, no data); a fixed 3.3 V LDO powers an STM32G030F6P6 MCU running on its internal oscillator, which reads a Sensirion SHT4x sensor over I2C. The same I2C bus is exposed on a Qwiic/STEMMA-QT connector; readout/logging is over a 4-pin UART header; programming/debug is over a 4-pin SWD header. One user LED and one power LED. Prototype quantity 5, JLC economy PCBA, meant to be ordered as-is.

2. Interfaces

- USB-C receptacle, POWER ONLY: 5 V VBUS input; 5.1 k pull-downs on CC1 and CC2 (so any USB-C source or a legacy A-to-C cable enables VBUS); D+/D- unconnected; no USB data.
- SWD header: 4-pin, 0.1 in pitch. ASSUMED: pins GND, 3V3, SWDIO, SWCLK (exact order per architect; brief states only "4-pin SWD header (0.1 in)").
- UART header: 4-pin, 0.1 in pitch: GND, 3V3, TX, RX - for readout/logging. ASSUMED: TX/RX are named from the MCU's perspective (TX = MCU transmit).
- Qwiic/STEMMA-QT connector: JST SH 1.0 mm 4-pin, GND/3V3/SDA/SCL, sharing the sensor I2C bus. I2C pull-ups required on the board (values per architect).

- User LED: one, driven from a GPIO through a series resistor.
- Power LED: one (which rail it indicates - VBUS or 3V3 - is per architect; the brief does not say).
- NRST push button (momentary reset).
- BOOT0: handled (method - strap, option bits, or jumper - per architect; the brief requires only that it is handled).
- Four M2 mounting holes, CONDITIONAL: only if they fit without hurting the layout (see section 5).

3. Power

- Input: USB-C VBUS, 5 V nominal, device draws < 1 A total. No battery, no charging, no mains.
- VBUS input protection required (product scope, brief-stated): TVS, resettable fuse or equivalent; reverse/inrush handling as the architect sees fit - the brief delegates the exact topology.
- 3.3 V rail: fixed LDO, ≥ 300 mA. JLC Basic/Preferred part preferred; AMS1117-3.3 class acceptable. Proper input/output capacitors per the chosen part's datasheet.
- Rail budget (GUESSES - to be replaced by the architect's numbers):
 - STM32G030F6P6 on internal oscillator: ~ 5 -15 mA (guess)
 - SHT4x: < 1 mA average, single-digit mA peaks during measurement (guess)
 - LEDs: ~ 2 -5 mA total (guess)
 - Qwiic downstream devices: not budgeted by the brief - see open question 2 (recommended default: reserve 100 mA).
 - On-board total is tens of mA; the brief's ≥ 300 mA LDO floor leaves the headroom for Qwiic loads.

4. Environment

Not stated in the brief. ASSUMED: indoor, room-ambient use (roughly 0-40 C), bare board, no ingress or vibration requirements (see open question 1). The one stated thermal/environmental requirement is layout-level: the SHT4x gets a thermal isolation slot/cutout, or at minimum is kept away from LDO and MCU heat, so self-heating does not corrupt the measurement.

5. Size & mounting

- Outline: "whatever an honest layout wants, roughly 35 x 25 mm or smaller". This is a SOFT preference - no HARD cap. It must NOT bind at P5 board_init; the layout earns the final outline, and ~ 35 x 25 mm is a target to aim for, not a box to squeeze into (see open question 5).
- Mounting: four M2 mounting holes, CONDITIONAL - "if they fit without hurting the layout". The brief itself allows dropping or reducing them in favor of layout honesty; record that decision if taken.
- Height: no limit stated.

6. Quantity & budget

- Quantity: 5 boards, prototype run, assembled.
- Unit cost: no target stated (see open question 3). Cost posture from the brief: JLC economy PCBA; prefer JLC Basic parts; Extended parts only where the brief names the part (MCU, sensor, connectors).

7. Assembly

- JLC economy PCBA for the SMD parts.
- ASSUMED: single-sided SMD assembly (economy PCBA norm, and this part count does not need the second side).
- Through-hole parts: economy PCBA assembles SMD only, so the two 0.1 in headers are ASSUMED to ship unpopulated for owner hand-soldering (see open question 4).
- ASSUMED: NRST push button and the USB-C / JST SH connectors are SMD, JLC-assemblable variants where stock allows, so they ride the PCBA run.
- Board: 2 layers, 1.6 mm FR-4, HASL, green (brief-stated).

8. Compliance/safety flags

Each flag checked against the brief; none applies:

- Mains voltage: NOT APPLICABLE - brief states "no mains"; input is USB 5 V.
- Battery: NOT APPLICABLE - brief states "no battery"; no charging circuitry anywhere in scope.
- Motors: NOT APPLICABLE - no motor or actuator in the brief's scope.
- >30 V: NOT APPLICABLE - the highest voltage on the board is USB VBUS at 5 V nominal.
- High current (>3 A): NOT APPLICABLE - the brief describes a USB 5 V, < 1 A device.
- RF transmit: NOT APPLICABLE - brief states "no RF"; no radio, and no USB data either ("D+/D- unconnected").

9. Open questions

Closed-form, each with a recommended default. Unattended run: the orchestrator answers these on the owner's behalf and records the answers.

1. Operating environment: is indoor, room-ambient use (about 0-40 C, bare board, no ingress/vibration requirements) correct? Recommended default: YES. 2. Qwiic 3.3 V budget: how much current may external Qwiic/STEMMA-QT devices draw from the 3.3 V rail? Recommended default: reserve 100 mA (fits under the ≥ 300 mA LDO floor together with the on-board load). 3. Unit cost: is there a hard per-board cost cap? Recommended default: NO hard cap - minimize via economy PCBA and Basic parts as the brief already directs. 4. Header population: should the two 0.1 in headers (SWD, UART) ship unpopulated for hand-soldering, given economy PCBA assembles SMD only? Recommended default: YES - footprints and holes present, headers not populated at assembly. 5. Size confirmation: is $\sim 35 \times 25$ mm truly a soft preference (the layout may exceed it if honesty demands), not an enclosure-driven hard cap? Recommended default: SOFT preference, per the brief's own wording ("whatever an honest layout wants").

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	unattended default: environment = indoor bench/room ambient, 0-40 C operating, bare board, no ingress or vibration rating	Q1 of requirements section 9; brief names no environment. Most conservative common-practice choice for a USB-powered indoor sensor node; parts will still be picked at industrial -40..85 C grade where that is the stocked option, so the assumption cannot make the board wrong.
P0	unattended default: Qwiic/STEMMA-QT down-stream budget = 100 mA reserved on the 3V3 rail	Q2; brief does not budget it. 100 mA covers typical Qwiic sensor/display loads and still fits inside the brief's ≥ 300 mA LDO floor with margin for the on-board tens of mA, so it sizes the LDO and its thermals honestly without inflating the part.
P0	unattended default: no hard per-board cost cap; minimize cost via JLC economy PCBA and Basic parts as the brief already directs	Q3; brief states no budget. Cost is already bounded by the brief's own Basic-parts-preferred rule; inventing a cap could force a worse engineering choice.
P0	unattended default: the two 0.1 in headers (SWD, UART) ship UN-POPULATED - plated through-holes only, hand-solderable; only SMD parts go to JLC economy PCBA	Q4; brief says 'JLC economy PCBA for the SMD parts'. JLC economy assembly is SMT-only, so THT headers are BOM 'do not populate / customer supplied'. Standard practice for dev/sensor boards and it keeps the assembly quote in economy tier.
P0	unattended default: $\sim 35 \times 25$ mm is a SOFT preference, not an enclosure-driven hard cap; no HARD outline cap binds at P5	Q5; the brief says 'whatever an honest layout wants, roughly 35×25 mm or smaller' and names no enclosure. Confirms requirements.md section 5; placement must not optimize to fit an unearned number (the bb-buck failure mode).
P1	unattended default: LDO = AMS1117-3.3 class in SOT-223 (JLC Basic, C6186); governing thermal design point is the brief's literal ≥ 300 mA rated case (0.51 W), not the ~ 150 mA realistic load	research/ldo.md escalated the SOT-223-vs-SOT-23-5 tradeoff; research/power.md resolved it with numbers: AMS1117 T_j 71-86 C at 40 C ambient and ≤ 115 C even in the 500 mA abuse case, while AP2112K-class SOT-23-5 (θ_{JA} 184 C/W) hits T_j 134 C at the rated case with no copper fix available. The Qwiic port is an unfenced 3V3 output, so the rated case is reachable in the field. Basic part, 1.1M stock, \$0.20.
P1	unattended default: a sub-6 V TVS clamping voltage is NOT a requirement; SMAJ5.0A/SMF5.0A-class (standoff 5.0-5.5 V, $V_c \sim 9.2$ V at I_{pp}) is accepted	research/protection.md flagged that no passive TVS clamps under 6 V. Nothing on VBUS is 6 V-rated: the only VBUS consumers are the LDO (AMS1117 V_{in} abs max 15 V), the input caps (16 V+ parts) and the PTC (16 V). Everything at 3.3 V sits behind the LDO. A ~ 9.2 V clamp during a surge is therefore inside every VBUS part's rating - the original < 6 V framing was an over-tight derived spec, not a real one.

Phase	What	Why
P1	unattended default: NO series reverse-polarity element (no Schottky, no P-FET ideal diode) on VBUS	Two independent reasons, both recorded in research: (a) a compliant USB-C cable cannot present reverse polarity (protection.md), and (b) research/power.md computed the end-to-end dropout budget and found a series Schottky's 125-190 mV would consume the entire AMS1117 headroom at the 4.75 V source corner (+0.11 V worst-corner margin as designed). The unidirectional TVS already crowbars a reverse fault. Adding it would make the board worse, not safer.
P1	unattended default: USB-C receptacle = 16-pin power-only part whose THT shield/mounting legs are accepted; JLC assembles it in the Economic PCBA tier	research/connectors.md correctly flagged that every 16P part in this class has THT shield legs and asked whether economy PCBA can place them. Verified directly on the JLCPCB part page for C165948 (TYPE-C-31-M-12): Assembly Process = SMT, Mount Type = Surface Mount Right Angle, and the part is listed as supported by BOTH Economic and Standard assembly. So the legs are not an assembly blocker. Part is JLC Extended - permitted, the brief names the connector.
P1	unattended default: SHT4x on-chip heater (75 mA peak) IS budgeted in the rail tree at <=10% duty	research/power.md's call, accepted: including it costs nothing (peak +3V3 still 185 mA, 240 mA with 30% headroom, under the 300 mA LDO floor and at 39% of the 500 mA CC-blind USB entitlement), while omitting it would turn a documented firmware feature into a hardware ECO later.
P1	unattended default: NRST tactile button - prefer the JLC Basic 5.1x5.1 mm part over the brief's smaller 3x2.5/4.2x3.2 mm size class if the layout has room; size class is a preference, Basic status and stock depth are the tiebreak	research/connectors.md Q2: the only Basic button is 5.1x5.1 mm (C318884, 962k stock, \$0.02); the closest-size alternative carries a documented lib_pull symbol gotcha in this repo's own LEARNINGS. The brief never fixed the button size, and the board size is soft, so paying ~2.9 mm of edge for a Basic, gotcha-free part is the conservative trade. Architect may overrule at P2 if placement suffers.
P1	unattended default: LED colours - user LED green, power LED red; both 0603, series resistors sized from the measured Vf band per colour	research/connectors.md Q3 left the assignment open. Red is the conventional power indicator and is the JLC Basic option with the low Vf band (1.8-2.4 V), which gives the largest resistor and the lowest current draw for an always-on LED. Green for the user LED is the conventional 'activity' colour; it is Extended in the KT-0603 family, which is acceptable at one extra unique part, and its 2.6-3.1 V Vf is fine from 3.3 V at a few mA.
P2	research task block-mcu-1 closed over 1 unverified draft record(s): mcu-boot-strap-only-when-option-bytes-consult-pin	-accept-drafts: the slot ships without this knowledge

Phase	What	Why
P2	unattended default: BOOT0 gets a populated 10k pull-down on PA14 (pin 19, shared with SWCLK) - the board does NOT rely on the unverified factory-option-byte claim	The research record mcu-boot-strap-only-when-option-bytes-consult-pin was REFUTED by a fresh second reader: its load-bearing claim (factory 0xDFFFE1AA / nBOOT_SEL=1 makes the BOOT0 pin ignored) rests solely on a community.st.com forum page, and RM0454/AN2606 could not be acquired - st.com times out from this container (verified twice by curl, once by an agent). Rather than design on unverified knowledge, the board is made correct under EITHER option-byte state: a 10k pull-down guarantees BOOT0=0 at reset so the part boots main flash even if nBOOT_SEL=0. Cost is one 0402 resistor; 330 uA load on SWCLK is negligible to any debugger's push-pull driver, and the datasheet-confirmed internal pull-down on PA14 in debug mode already pulls the same way. This closes the gap without the source.
P2	unattended default: I2C pull-ups are 1.5 kOhm (not the 2.2 kOhm the P1 fragment proposed)	The draft record sht4x-i2c-pullup-bus-cap was refuted for recommending 2.2k over its own declared 150-200 pF bus budget: UM10204 Eq 1 at tr=300 ns gives $R_p(\max) = 354/C_b[\text{pF}]$ kOhm, i.e. 1.77k at 200 pF, so 2.2k needs $C_b \leq 161$ pF and misses the Fast-mode rise time on a cabled Qwiic bus. 1.5k holds the ceiling to 236 pF and clears both floors (967 Ohm from UM10204 Table 10 at 3.3 V/3 mA, 390 Ohm from the SHT4x datasheet). Value independently recomputed by a second reader. architecture/constraints.json and the P4 schematic must use 1.5k.

Sheet plan

sheets - g0-sense hierarchical plan (P2)

Small board: root + TWO child sheets. One sheet would also fit, but the power/main split gives P6 its natural placement groups (power entry corridor vs logic/sensor), keeps the Type-C + protection cluster reviewable in one frame, and costs nothing. Refdes are unique ACROSS sheets by disjoint ranges; every range includes a #PWR base.

Root: g0-sense.kicad_sch

Holds the two sheet symbols only. Global power nets (power symbols, bare netlist names): VBUS, +5V, +3V3, GND. Power symbols make these names bare in the final netlist; all constraints.json net references use the bare forms.

Sheet 1: power.kicad_sch (blocks B1 usb-input+protection, B2 ldo-3v3)

- Contents: J1 USB-C receptacle; R1, R2 (5.1k Rd); D1 (TVS); F1 (PTC); C1 (100 nF VBUS); U1 (AMS1117-3.3); C2 (10 uF X5R, +5V at VIN); C3 (22 uF tantalum, +3V3); D2 (power LED red) + R3 (620 Ohm).
- Interface nets (hier pins -> placement groups at P6): +3V3 (out), GND. VBUS and +5V stay sheet-local (they leave via power symbols, so their netlist names are still bare VBUS / +5V).

- Sheet-local labels: CC1, CC2 (final names /power/CC1, /power/CC2).
- Refdes ranges: J1; U1; F1; D1-D9; R1-R9; C1-C9; SW none; #PWR pwr_base=100 (#PWR0100-#PWR0199).

Sheet 2: main.kicad.sch (blocks B3 mcu, B4 sht4x + I2C/Qwiic, headers)

- Contents: U2 (STM32G030F6P6); U3 (SHT40-AD1B class); C10 (100 nF U2), C11 (4.7 uF U2), C12 (100 nF NRST), C13 (100 nF U3); R10, R11 (1.5k I2C pull-ups), R12 (220 Ohm user LED), R13 (10k BOOT0 pull-down); D10 (user LED green); SW1 (NRST button); J2 (Qwiic JST SH); J3 (SWD 1x4 DNP); J4 (UART 1x4 DNP); H1-H4 (M2 mounting holes, conditional).
- Interface nets: +3V3 (in), GND.
- Sheet-local labels (final names /main/<label>): SDA, SCL, SWDIO, SWCLK, UART_TX, UART_RX, NRST, LED_USER.
- Refdes ranges: U2-U9; J2-J9; D10-D19; R10-R19; C10-C19; SW1-SW9; H1-H9; #PWR pwr_base=200 (#PWR0200-#PWR0299).

Interface-net contract (P4 must produce exactly these)

```
| Net (netlist form) | Class | From | To |
|----|----|----|----|
| VBUS | power (bare) | J1 | D1, F1, C1 |
| +5V | power (bare) | F1 | U1 VIN, C2 |
| +3V3 | power (bare) | U1 VOUT | C3, D2/R3, U2, U3, R10, R11, J2, J3.2, J4.2 |
| GND | power (bare) | all | all + J1 shield |
| /power/CC1, /power/CC2 | signal | J1 CC pins | R1, R2 |
| /main/SDA, /main/SCL | signal | U2 PA10/PA9 | U3, J2, R10/R11 |
| /main/SWDIO, /main/SWCLK | signal | U2 PA13/PA14 | J3.3/J3.4; SWCLK also R13
-> GND |
| /main/UART_TX, /main/UART_RX | signal | U2 PA2/PA3 | J4.3/J4.4 |
| /main/NRST | signal | U2 pin 6 | C12, SW1 |
| /main/LED_USER | signal | U2 PA5 | R12 -> D10 |
```

Header pinouts (canonical): J3 SWD = 1:GND, 2:3V3, 3:SWDIO, 4:SWCLK. J4 UART = 1:GND, 2:3V3, 3:TX (MCU transmit), 4:RX (MCU receive). Pin 1 marked on silk on both; no vendor standard exists for a 4-pin SWD header, so the silk labels are the contract.

Full architecture narrative (not inlined; see the artifact index): [architecture/blocks.md](#), [architecture/decisions.md](#), [architecture/power_tree.md](#), [architecture/stackup.md](#).

5 Schematic

Pending — produced at P4.

6 Layout

Pending — produced at P6.

7 Verification

Pending — produced at P8.

8 DFM and Fabrication

Pending — produced at P9.

9 Run Record (Appendix)

Phase digests

P0

P0 Intake digest - g0-sense (2026-08-27)

- Artifact: 'requirements.md' (9 sections), 'check_requirements.py' PASS (0 violations).
- Mode: NO mode token in the brief -> mode = none, design normally. Scope is product per the brief's own words, so absent protection/filtering/connectors/thermal IS a reviewer finding downstream.
- Geometry: ~35 x 25 mm recorded SOFT (no HARD cap) - must not bind at P5 board_init; four M2 holes conditional on the layout.
- Safety (section 8): mains / battery / motors / >30 V / >3 A / RF transmit all

NOT APPLICABLE with brief evidence (USB-C 5 V power-only, <1 A). No safety unknown to escalate; nothing PROVISIONAL on safety grounds.

- 5 OPEN questions answered as unattended defaults and recorded as decisions: environment indoor 0-40 C; Qwiic budget 100 mA; no hard cost cap; 0.1 in headers ship unpopulated (economy PCBA is SMT-only); size stays soft.
- Spawn: requirements-analyst @ fable/high (session effort xhigh).
- Next: P1 research roster (power architect, component scout, interface spec).

P1

P1 Research digest - g0-sense (2026-08-27)

Roster (8 agents): 4x research-component-scout (ldo, protection, connectors, mcu-sensor), research-interface-spec (usbc-power-sink), 2x research-reference-design (sht4x, stm32g030-minimal), research-power-architect.

Load-bearing findings

- USB-C CC-blind sink (fixed 5.1k Rd on CC1 AND CC2, independent - never one shared resistor) is entitled to DEFAULT USB power only: budget <= 500 mA. Sink capacitance at the receptacle <= 10 uF (TC2.0 Table 4-3).
- Rail tree: VBUS -> TVS -> PTC -> +5V -> LDO -> +3V3. +3V3 peak 185 mA (incl. 100 mA Qwiic reserve + 75 mA SHT4x heater), 240 mA with 30% headroom. VBUS worst draw 196 mA = 39% of entitlement.
- LDO: AMS1117-3.3 SOT-223 (C6186, Basic) beats SOT-23-5 low-Iq parts on the only axis that decides: Tj 71-86 C vs 134 C at 40 C ambient, 0.51 W case. Needs a 22 uF TANTALUM-class output cap (not plain ceramic) - carry to P3/P4. Dropout closes at the 4.75 V corner with +0.11 V worst-case margin.
- STM32G030F6P6 TSSOP-20: VDD/VDDA bonded to pin 4, VSS/VSSA pin 5, no separate VREF+. BOOT0 needs NO strap - factory nBOOT_SEL=1/nBOOT0=1 always boots main flash, and BOOT0 shares pin 19 with SWCLK anyway. NRST needs only 100 nF + the button (internal pull-up 40k typ). Pin map: SWD PA13/PA14, USART2 PA2/PA3, I2C1 PA9/PA10 (5 V tolerant, Fm+), LED PA5. 9 of 20 pins used, no conflicts.
- SHT4x: 100 nF at VDD, 2.2k I2C pull-ups (sized for a 0.5 m Qwiic leg), copper conduction is the dominant self-heating path -> distance + necked traces + no pour under the part + a milled slot. NO BOARD WASH (Sensirion forbids it) -> must appear in the fab/assembly notes. Fixed I2C address per variant.

- MCU + sensor are both JLC EXTENDED (no Basic exists in either family):
STM32G030F6P6 C724040 \$1.37, SHT40-AD1B-R2 C2909890 \$1.90 (addr 0x44).

7 OPEN questions answered and recorded as decisions (LDO design point + part;
TVS clamp ceiling dropped as an over-tight derived spec; no reverse-polarity
element; USB-C THT legs verified OK in JLC Economic tier; heater budgeted;
button size vs Basic; LED colours). Gates: none due at P1.

P2 (*no digest recorded*)

Gate attempt history

(*no entries*)

Issues

(*no entries*)

Human checkpoints

Id	Phase	Status	When	Note
1	P2	not reached		
2	P4	not reached		
3	P6	not reached		
4	P8	not reached		
5	P10	not reached		

10 Artifact Index

File	Size	Note
state.json	19,847 B	
brief/brief.md	1,634 B	
requirements.md	6,653 B	
architecture/blocks.md	11,799 B	
architecture/constraints.json	8,406 B	
architecture/decisions.md	7,408 B	
architecture/power_tree.md	4,040 B	
architecture/sheets.md	2,964 B	
architecture/stackup.md	2,679 B	